



Class Work(04.08.26)

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QUESTIONS

Q1

First Repeated
5 marks

**Q2**

Duplicate Elements
5 marks

**Q3**

Compare Lists
5 marks

**Q4**

Duplicate Tuples
5 marks

**Q5**

Pair Sum
5 marks

**Q6**

Unique Characters
5 marks

**Q7**

Unique Words



Q1 First Repeated

5 marks

Write a Python program to find the first repeated element in a list using a set.

Input:

10 20 30 20 40

Output:

20

Your Code

```
1 s=set()  
2 for x in map(int,input().split()):  
3     if x in s: print(x); break  
4     s.add(x)
```

Input

10 20 30 20 40

Output

20

Visible Test Cases

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QUESTIONS

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5 marks**Q5**Pair Sum
5 marks**Q6**Unique Characters
5 marks**Q7**

Unique Words



Q2 Duplicate Elements

5 marks

Write a Python program to find all duplicate elements in a list using a set.

Input:

10 20 30 20 10

Output:

{10,20}

Your Code

```
1 nums = list(map(int, input().split()))
2
3 seen = set()
4 duplicates = set()
5
6 for x in nums:
7     if x in seen:
8         duplicates.add(x)
9     else:
10        seen.add(x)
11
12 print(duplicates)
```

Input

10 20 30 20 10

Output

{10, 20}

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5 marks

Q3 Compare Lists

5 marks

Write a Python program to check whether two lists contain the same unique elements.

Input:

List1:1 2 2 3

List2:3 2 1

ouput:

True

Your Code

```
1 a=list(map(int,input().split())); b=list(map(int,
2 print(set(a)==set(b)))
```

Input

```
1 2 2 3
3 2 1
```

Output

```
True
```

Visible Test Cases



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5 marks



Q5

Pair Sum
5 marks



Q6

Unique Characters
5 marks



Q7



Q4 Duplicate Tuples

5 marks

Write a Python program to remove duplicate tuples from a list using a set.

Input:

[(1,2),(1,2),(3,4)]

Output:

[(1,2),(3,4)]

Your Code

```
1 tuples_list = [(1,2), (1,2), (3,4)]  
2  
3 result = list(set(tuples_list))  
4 print(result)
```

Input

stdin input

Output

[(1, 2), (3, 4)]

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5 marks**Q7**Unique Words
5 marks

Q5 Pair Sum

5 marks

Write a Python program to find all pairs of elements whose sum equals a given number.

Input:

List:2 4 6 8 10

Target=12

ouput:

(4,8),(2,10)

Your Code

```
1 a=list(map(int,input().split()))
2 t=int(input())
3 s=set();out=[]
4 for x in a:
5     y=t-x
6     if y in s: out.append((y,x))
7     s.add(x)
8 print(",".join(map(str,sorted(set(out)))))
```

Input

2 4 6 8 10
12

Output

(2, 10), (4, 8)

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Pair Sum

5 marks

**Q6**

Unique Characters

5 marks

**Q7**

Unique Words

5 marks



Q6 Unique Characters

5 marks

Write a Python program to count the number of unique characters in a string.

Input:

Programming

Output:

8

Your Code

```
1 s = input().strip()
2 print(len(set(s)))
```

Input

programming

Output

8

Visible Test Cases

Test Case 1



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5 marks

Q7 Unique Words

5 marks

Write a Python program to display the unique words in a sentence using a set.

Input:

python is easy python

{'python','is','easy'}

Your Code

```
1 s = input().strip()
2 words = set(s.split())
3 print(words)
```

Input

python is easy python

Output

```
{'python', 'easy',
'is'}
```

Visible Test Cases

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 Unique Words
5 marks


Q8 Exactly One Set

5 marks

Write a Python program to find the elements that appear in exactly one of three sets.

Input:

A={1,2,3}

B={3,4,5}

C={5,6,7}

ouput:

{1,2,4,6,7}

Your Code

```
1 A = {1, 2, 3}
2 B = {3, 4, 5}
3 C = {5, 6, 7}
4
5 result = (A | B | C) - ((A & B) | (A & C) | (B & C))
6
7 print(result)
```

Input

stdin input

Output

{1, 2, 4, 6, 7}



Program terminated.

Your Code

```
1 s=set()
2 while True:
3     c=int(input("Enter your choice: "))
4     if c==1:
5         s.add(input("Enter student name: ").strip())
6     elif c==2:
7         n=input("Enter student name to remove: ")
8         print("Student removed successfully." if n in s else "Student not found.")
9     elif c==3:
10        n=input("Enter student name to search: ")
11        print("Student found." if n in s else "Student not found.")
12    elif c==4:
13        print("Students:",s)
14    elif c==5:
15        print("Total number of students:",len(s))
16    elif c==6:
17        print("Program terminated."); break
18    else:
19        print("Invalid choice!")
```

Input

3
Ajay

Output

```
Security Error:
Disallowed code pattern
detected.
```



Submitted





5 marks

Q9

Menu Driven
5 marks



Q10

Student Database
5 marks



Enter your choice: 4

Sample Output

Symmetric Difference: {1, 2, 5, 6}

Sample Input

Enter your choice: 5

Sample Output

Program terminated.

10/10 answered

Your Code

```
1 A = set(map(int, input("Enter the first set elements: ").split()))
2 B = set(map(int, input("Enter the second set elements: ").split()))
3
4 while True:
5     print("\n1.Union 2.Intersection 3.A-B 4.SymDiff 5.Exit")
6     c = input("Enter your choice: ")
7     if c == '1': print("Union:", A | B)
8     elif c == '2': print("Intersection:", A & B)
9     elif c == '3': print("Difference:", A - B)
10    elif c == '4': print("Symmetric Difference:", A ^ B)
11    elif c == '5': print("Program terminated.")
12    else: print("Invalid choice")
```

Input

```
1 2 3 4
3 4 5 6
5
```

Output

```
Enter the first set
elements: Enter the
second set elements:
1.Union
2.Intersection 3.A-B
4.SymDiff 5.Exit
```



Submitted